MIL-T-12333C(EL)
28 February 1969

SUPERSEDING
MIL-T-12333B(EL)
10 March 1966

MILITARY SPECIFICATION

TEST SET, CRYSTAL UNIT, QUARTZ TS-710( )/TSM

1. SCOPE.

1.1 This specification covers one type of crystal test set, designated Test Set, Crystal Unit, Quartz TS-710( )/TSM.

2. APPLICABLE DOCUMENTS.

2.1 Documents. The following documents of the issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping.
PPP-B-636	Box, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	Paperboard, Wrapping Cushioning.
PPP-S-760	Strapping, Nonmetallic, (and Connectors).
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 6625

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MIL-E-75	Electron Tube Packing, Packaging, and Container Marking for General Specification for.
MIL-R-94	Resistor, Variable, Composition, General Specification for.
MIL-P-116	Preservation, Methods of.
MIL-C-3098	Crystal Units, Quartz.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-B-43291	Boxes; Fiberboard; Corrugated, Double-Wall, Weather Resistant.

STANDARDS

MILITARY

MS-35214	Screw, Machine - Pan Head.
MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads; 40 in. x 48 in. 4-Way Partial and 4-Way Pallets.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-781	Reliability Tests Exponential Distribution.
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester.
SC-DL-346835	Test Set, Crystal Unit, Quartz TS-710( )/TSM.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS.

3.1 Description. The Test Set, Crystal Unit, Quartz TS-710( )/TSM measures the series resonant and antiresonant equivalent resistance of quartz crystals in the range of 10 kHz to 1,100 kHz for conformance to the military specification requirements of that specified crystal.* In addition to the basic unit, this equipment includes the following accessories:

(a) Crystal adaptor assemblies.

(b) Calibration resistor assemblies.

(c) Calibration potentiometer assemblies.

(d) Calibration chart.

(e) Cord CG-433( )/U (5 ft + 1 in - 0 in).

* The equipment shall operate from a 115 and 230 Vac ±10 percent, 50 to 400 Hz, single phase source.

3.2 First article samples. The contractor shall furnish three (3) complete first article samples for approval provided the invitation for bid and the contract requires first article samples.

3.3 Construction. The equipment shall be constructed in accordance with SC-DL-346835 and the requirements specified herein.

3.3.1 Model. A model shall be furnished to the contractor as required to meet requirements of 3.7.5.

3.4 Parts, materials and processes-general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection of requirements therein, shall apply.

3.5 Finish, protective. The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware; such as screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be final Film E, semigloss, light gray enamel, conforming to MIL-F-14072.

3.6 Marking. Marking shall conform to MIL-M-13231.

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3.6.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings shall be readily visible with a minimum disassembly of the equipment.

3.6.2 Serial numbers. Serial numbers are required on each nomenclatured and type designated item, except cords and cables.

3.7 Electrical.

3.7.1 Frequency range. The crystal impedance meter shall oscillate throughout the frequency range of 10 kHz to 1100 kHz. The accuracy of the signal generated shall not deviate by more than ±10 percent from the indicated nominal values, when tested in accordance with 4.7.1.

3.7.2 Power supply. The equipment shall operate from a 115/230V ±10 percent, 50/400 Hz, single phase source when tested in accordance with 4.7.2.

3.7.3 Power vs resistance.

3.7.3.1 0.1 mW operation (high resistance). The power vs resistance characteristic shall be within ±50 percent or 0.1 mW whichever is greater, of the values listed below when tested in accordance with 4.7.3.1.

Resistance (ohms)	Power (mW)
7 k	2.5
20 k	1.5
75 k	0.10 (initial setting)
100 k	0.03

3.7.3.2 0.1 mW operation (low resistance). The power vs resistance characteristic shall be within ±50 percent or 0.1 mW whichever is greater, of the values listed below when tested in accordance with 4.7.3.2.

Resistance (ohms)	Power(mW)
56	0.10
220	0.16
390	0.10 (initial setting)
560	0.07

3.7.3.3 2.0 mW operation. The power vs resistance characteristic shall be within ±50 percent, or 2.0 mW, whichever is greater, of the values listed below when tested in accordance with 4.7.3.3.

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Resistance (ohms)	Power (mW)
100	5
1,000	15
7,000	2 (initial setting)
10,000	1

3.7.4 Voltmeter accuracy. The voltmeter readings shall be accurate to within ±10 percent when tested in accordance with 4.7.4.

3.7.5 Resistance determination. The crystal impedance meter shall be capable of determining the equivalent resonant, or anti-resonant resistance of crystals manufactured in accordance with MIL-C-3098 in the frequency range of 10 kHz to 1100 kHz, at drive levels of 0.1 mW, 0.5 mW, and 2 mW, to within an accuracy of ±10 percent when tested in accordance with 4.7.5.

3.7.6 Ohmmeter accuracy. The ohmmeter readings shall be accurate to within ±5 percent for all readings above half-scale on all resistance ranges when tested in accordance with 4.7.6.

3.7.7 Load capacitance. The capacitance of the load capacity shall vary from less than 20 pF to more than 100 pF for any setting of the dial. The value of capacitance, as determined from the serialized calibration chart, shall be accurate to within ±0.2 pF up to a dial setting of 30, and to within ±0.5 pF for dial settings above 30 when tested in accordance with 4.7.7.

3.7.8 Capacitance across terminating resistance. The maximum capacitance in parallel with the terminating resistance shall be 50 pF, when tested in accordance with 4.7.8.

3.7.9 Accessories.

3.7.9.1 Adapters. Two adapters (one for type HC-5/U and one for octal base crystal holders) are furnished. The adapter for the octal base crystal holder shall be wired so that pins 2, 3, and 4 are connected to one another, and pins 5, 6, and 7 are connected to one another, with pin one connected to ground. They shall be tested in accordance with 4.7.9.1.

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3.7.9.2 Calibration resistor assemblies. Fourteen 1/2 watt, ±1 percent, metalized film resistors, each mounted on a suitable base which can be plugged into the crystal, or ohmmeter, socket, are furnished in the following values: 1,000, 2,000, 3,000, 4,000, 5,000, 6,000, 7,000, 8,000, 10,000, 20,000, 50,000, 75,000, 100,000, 125,000 ohms. Each resistor assembly shall be marked with its resistance value and this value shall not deviate by more than 1 percent when tested in accordance with 4.7.9.2.

3.7.9.3 Calibration potentiometer assemblies. One each 5,000, 50,000 and 500,000 ohm composition potentiometers are furnished in accordance with MIL-R-94. Suitable plugs are attached for inserting into the crystal or ohmmeter socket. Each potentiometer assembly shall be marked with its maximum resistance. Its maximum value shall not deviate by more than ±10 percent and shall have continuity from one end of rotation to the other when tested in accordance with 4.7.9.3.

3.8 Service conditions. The Test Set, Crystal Unit, Quartz TS-710( )/TSM shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified herein when subjected to the following conditions.

3.8.1 Fungus. The equipment, including accessories, shall show no evidence of viable fungus and shall meet the requirements of 3.7.1 when tested in accordance with 4.8.1.

3.8.2 Low temperature. The equipment, including accessories shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.1 at times specified when tested in accordance with 4.8.2.

3.8.3 High temperature. The equipment, including accessories shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.1 at times specified when tested in accordance with 4.8.3.

3.8.4 Humidity. The equipment, including accessories shall exhibit no physical damage of parts and materiel, such as corrosion, rust, blistering, swelling, deterioration, etc., or electrical degradation, and shall meet the requirements of 3.7.1 at times specified, when tested in accordance with 4.8.4.

3.8.5 Altitude. The equipment, including accessories shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.1 at times specified when tested in accordance with 4.8.5.

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3.8.6 Shock, bench-handling. The equipment, less accessories, shall show no evidence of physical damage except minor surface abrasion and shall meet the requirements of 3.7.1 when tested in accordance with 4.8.6.

3.8.7 Vibration. The equipment, including accessories, shall show no evidence of breakage or loosening of parts, and shall meet the requirements of 3.7.1 when tested in accordance with 4.8.7.

3.8.8 Electromagnetic interference. The equipment shall comply with CE03, CS01, CS02, CS06, RE02 (.014 to 400 MHz) and RS03 (.014 to 400 MHz) interference and susceptibility requirements for Class 1C equipments of MIL-STD-461 when tested in accordance with 4.8.8.

3.9 Visual and mechanical. The complete equipment shall meet the requirements of MIL-STD-252 as specified in 4.9.

3.10 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the equipment. Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc. (see 4.10).

3.11 Burn-in and debugging. Each equipment less accessories produced shall be subjected to the burn-in and debugging test of 4.11 after bounce preconditioning and prior to Group A inspection. The equipment shall be monitored for frequency range per 3.7.1 and 4.7.1 during period of test.

3.12 Bounce preconditioning. Each equipment, including accessories, produced shall be subjected to the bounce preconditioning test of 4.6 prior to Group A inspection. After bounce preconditioning, the equipment, including accessories, shall not be repaired, aligned, cleaned or otherwise changed prior to performing Group A inspection.

3.13 Workmanship. The complete equipment shall be manufactured and assembled in accordance with the applicable portions of MIL-P-11268, as follows:

General requirements for plastic material and parts.
Wiring and cabling including:

Slack.
Protection.
Clearance.
Splicing.
Connectors, general.
Grounding, general.

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Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment.
General process for securing of parts.
Process of welding.
Connectors, general.
Mechanical operation of controls.
Tropicalization of material.
Lubricants.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (Does not include preparation for delivery) (see 4.3).

(b) Inspection covered by subsidiary documents (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables I, II and III respectively) on the three (3) samples of the completely assembled equipment. The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government production engineer. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.5
Marking	3.6
Workmanship	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government - at its discretion - may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all the Group A requirements and tests in Table I. Each lot shall be subjected to Government verification and this inspection, including sampling, shall conform to Table I and the ordinary inspection procedures of MIL-STD-105.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE I. Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL	
			Major	Minor
Visual and mechanical	3.9	4.9	1 dphu*	4 dphu
Electrical				
Frequency range	3.7.1	4.7.1		
Power supply	3.7.2	4.7.2		
Power vs resistance				
0.1 mW operation (high resistance)	3.7.3.1	4.7.3.1		
0.1 mW operation (low resistance)	3.7.3.2	4.7.3.2		
2.0 mW operation	3.7.3.3	4.7.3.3	1 dphu	
Voltmeter accuracy	3.7.4	4.7.4	for the	**
Resistance determination	3.7.5	4.7.5	group	
Ohmmeter accuracy	3.7.6	4.7.6		
Load capacity	3.7.7	4.7.7		
Capacitance across terminating resistance	3.7.8	4.7.8		

* Defects per hundred units.
** All electrical tests are major.

4.5.2 Group B inspection. This inspection, including sampling shall conform to Table II and to the special procedures for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection, and on samples selected from units that have been subjected to and meet the Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

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TABLE II. Group B inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.10	4.10	4 dphu*
Electrical			
Accessories			
Adapters	3.7.9.1	4.7.9.1	
Calibration resistor assemblies	3.7.9.2	4.7.9.2	1 dphu for the group
Calibration potentiometer assemblies	3.7.9.3	4.7.9.3	

*Defects per hundred units.

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met Group A inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for inspection.

4.5.3.1.1 Three (3) samples of complete equipment shall be selected at random, one (1) sample for each Group 1, Group 2 and 4, and Group 3 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. These samples shall constitute Group C requirement for first 100 units produced.

4.5.3.1.2 Thereafter, two (2) samples of complete equipment shall be selected at random, one (1) sample for each Group 2 and Group 3 in Table III from every 100 units and fraction thereof produced.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall be subject to and meet frequency range per 3.7.1 and 4.7.1.

4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials,

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as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. Group C inspection

Inspection	Req. Para.	Test Para.
Group 1		
Fungus	3.8.1	4.8.1
Group 2		
Low temperature	3.8.2	4.8.2
High temperature	3.8.3	4.8.3
Humidity	3.8.4	4.8.4
Group 3		
Altitude	3.8.5	4.8.5
Shock-bench handling	3.8.6	4.8.6
Vibration	3.8.7	4.8.7
Group 4		
Electromagnetic interference	3.8.8	4.8.8

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

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4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6 Bounce preconditioning. Each equipment, including accessories, produced shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.12.

4.7 Electrical.

4.7.1 Frequency range. Connect equipment output cord, CX-433( )/U, to an electric counter that has a frequency range of from 10 kHz to 1100 kHz inclusive. The electronic counter shall not deviate by more than ±1 count per time base division.

(a) Insert a 3000 ohm calibrating resistor in the crystal socket.

Set:

Termination resistor (ohms)	to 1000 ohms
A. R. switch	to "R"
Gain	to full counter-clockwise
Frequency range	to "10 - 25"
Tuning	to lowest indicated frequency
Termination resistor (Volts)	to "5.0"
Selector	to operate

(b) Advance gain until meter reads approximately half-scale.

(c) Read frequency on electronic counter.

(d) Repeat (a) through (c) with frequency range set to 25-70, 70-200, 200-450 and 450-1100 kHz respectively.

(e) Repeat (a) through (d) with tuning set to the middle position.

(f) Repeat (a) through (d) with tuning set to highest indicated frequency.

(g) The results shall meet the requirements of 3.7.1.

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4.7.2 Power supply. Connect equipment power cord, CX-112( )/U, to a power source capable of supplying ac voltages of 100 to 260 volts at frequencies of 50 to 400 Hz.

(a) Repeat 4.7.1 (and subparagraphs) with input voltages set to 103.5, 115 volts and 126.5 volts respectively at a frequency of 50 Hz.

(b) Repeat (a) with the power source frequency set at 400 Hz.

(c) Repeat (a) and (b) with the power source set at 207 volts, 230 volts, and 253 volts respectively.

(d) The results shall meet the requirements of 3.7.2.

4.7.3 Power vs resistance. Connect output cord, CX-433( )/U, to the electronic counter. A VTVM which is capable of operating in the range of 10 kHz to 1100 kHz is connected to the "EXT." VTVM jack. The voltage range of the VTVM shall be from 1 mV to 25 volts inclusive with an accuracy of ±2 percent of the full scale reading.

4.7.3.1 0 - .1 mW operation (high resistance).

(a) Insert a 75 k calibrating resistor in the crystal socket.

(b) Set:

AR switch	to	"R"
Termination resistor (ohms)	to	"10,000 OHMS"
Frequency range	to	"10 - 25"
Selector	to	"OPERATE"

(c) Adjust:

Tuning	to	10 kHz on electronic counter
Gain	to	.365 volts on the .5 V scale

This is the voltage setting for 0.1 mW dissipation in the 75 k calibration resistor. The gain control is not to be changed throughout the remainder of this measurement.

(d) Replace the 75 k calibration resistor with the 7 k, 20 k, and 100 k calibrating resistors respectively, and read the voltages in each case on the equipment meter. The dissipated power may be calculated from the following equation:

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$$P = \frac{E^2}{R_T^2} \cdot R_c$$

Where: P = power dissipated in Rc in watts
E = voltage across R_T in volts
R_T = terminating resistance in ohms
R_c = calibrating resistance in ohms

(e) Repeat (b) through (d) with frequency range set to 70 - 200 and Tuning adjusted to 100 kHz on the electronic counter. The results shall meet the requirements of 3.7.3.1.

4.7.3.2 0.1 mW operation (low resistance).

(a) Insert a 390 ohm, 1/2 watt ±1 percent metalized film resistor in the crystal socket.

(b) Set:

AR switch	to	"R"
Termination resistor (ohms)	to	"100"
Frequency range	to	"70 - 200"
Selector	to	"OPERATE"

(c) Adjust:

Tuning to 160 kHz on electronic counter.
Gain to 0.05 volt on VTVM (equipment meter cannot be used at this low voltage).

This is the setting for 0.1 mW dissipation in the 390 ohm resistor. The gain control is not to be changed throughout the remainder of this measurement.

(d) Replace the 390 ohm resistor with a 56, 220 and 560 ohms (1/2 watt, ±1 percent metalized film) resistor respectively, noting each voltage reading on the VTVM and calculating the corresponding power dissipated in each case as in 4.7.3.1(d).

(e) Repeat (b) through (d) with frequency range set to 200 - 450 and tuning adjusted to 330 kHz on the electronic counter. The results shall meet the requirements of 3.7.3.2.

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4.7.3.3 2.0 mW operation.

(a) Insert a 7 k calibration resistor in the crystal socket.

(b) Set:

AR switch	to	"R"
Termination resistor (ohms)	to	"1000"
Frequency range	to	"70 - 200"
Selector	to	"OPERATE"

(c) Adjust:

Tuning to 80 kHz on electronic counter.
Gain to .535 on 1.0V scale.

This is the setting for 2 mW dissipation in the 7 k resistor. The gain control is not to be changed throughout the remainder of this measurement.

(d) Replace the 7 k calibration resistor with a 100, 1000 and 10,000 ohm (1/2 watt ±1 percent metalized film) resistor respectively noting each voltage reading on the equipment meter and calculating the corresponding power dissipated in each case as in 4.7.3.1(d).

(e) Repeat (b) through (d) with the tuning adjusted to 450 kHz on the electronic counter. The results shall meet the requirements of 3.7.3.3.

4.7.4 Voltmeter accuracy. Connect as in 4.7.3 and insert the 50,000 calibrating potentiometer in the crystal socket.

(a) Set:

AR switch	to	"R"
Frequency range	to	"10 - 25"
Tuning	to	"10 kHz"
Termination resistor, volts full scale	to	"o.5"
Terminating resistor (ohms)	to	"10,000"

(b) Adjust gain control and calibrating potentiometer to give a reading of 20 percent of full scale on the panel meter. Note the corresponding reading on the VTVM. The results shall meet the requirements of 3.7.4.

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(c) Repeat (b) for readings of 40, 60, 80 and 100 percent of full scale respectively on the panel meter.

(d) Repeat (b) and (c) with termination resistor, volts full scale set to 1.0, 2.5, 5.0, and 10.0 volts respectively.

(e) Repeat (b), (c) and (d) with frequency range set to "450-1000" and tuning to "500".

4.7.5 Resistance determination. Connect as in 4.7.3.

(a) Set the following controls in accordance with the specifications of the crystal to be tested:

AR switch to "R".
Frequency range and tuning to approximate oscillating frequency of the crystal.
Load capacity to value indicated (if anti-resonant crystal).
Gain to full counter clockwise.
Termination resistor (ohms) to nearest value of the maximum equivalent resonant (or anti-resonant) resistance of the crystal.

(b) Insert in the crystal socket a fixed calibration resistor approximately equal to 70 percent of the maximum equivalent resistance of the crystal.

(c) Adjust tuning, as indicated on electronic counter, to the nominal crystal frequency.

(d) Adjust gain to that voltage reading which yields the proper power drive level for the crystal to be used in the test. This voltage is to be calculated from the following equation:

$$V_p = \sqrt{R_T^2 \; P/R_c}$$

Where: P = power dissipated in R_c in watts
V_p = voltage reading on meter
R_T = terminating resistance in ohms
R_c = calibrating resistance in ohms

It may be necessary to readjust gain and tuning several times to obtain the voltage of (c) and the frequency of (d). The gain setting shall not be changed throughout the remainder of this measurement.

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(e) Remove the calibration resistor and re-insert the crystal in the crystal socket. (If this is an anti-resonant crystal, the AR switch must be set to "A" prior to insertion of the crystal).

(f) Record voltmeter and frequency reading.

(g) Remove crystal and insert in crystal socket the smallest calibration potentiometer whose maximum resistance is above the maximum equivalent resistance of the crystal type.

(h) Set AR switch to "R".

(i) Adjust calibration potentiometer to the same voltage reading as in (f).

(j) Adjust tuning to same frequency reading as in (f).

(k) Repeat (e) through (j) until the voltage and frequency readings for the crystal and the calibration potentiometer are equal to each other.

(l) Remove calibrating potentiometer from crystal socket, being careful not to disturb its setting. Measure its resistance on the resistance bridge. The resistance bridge shall have a measurement range of from 1 ohm to 1 megohm with an accuracy of ±.1 percent.

(m) Repeat (a) through (l), using the Government furnished test set, TS-710( )/TSM, as a reference standard. The resistance value obtained in (l) shall meet the requirements of 3.7.5.

(n) Repeat (a) through (m) for all the crystal units listed below:

Crystal unit	Freq. (kHz)	Max. resistance (ohms)	Drive (mW)
CR-50A/U	16 - 30	100 K	0.1 mW
CR-38A/U	80 - 100	90 K	0.1 mW
CR-42A/U	100 - 200	4.5 K	2.0 mW
CR-25A/U	200 - 250	1.5 K	2.0 mW
CR-25A/U	250 - 350	2.0 K	2.0 mW
CR-47A/U	455 - 500	8.5 K	2.0 mW

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4.7.6 Ohmmeter accuracy. Measure the resistance of the resistors (1/2 watt, ±1 percent metalized film resistance) listed below on the ohmmeter scales indicated and compare with the values obtained on a resistance bridge. They shall meet the requirements of 3.7.6.

Resistance (ohms)	Ohmmeter scale
200	RX1
800	RX1
2,000	RX10
8,000	RX10
20,000	RX100
80,000	RX100

4.7.7 Load capacity.

(a) Use the connecting jig, constructed in accordance with figure (1). Connect the crystal impedance meter to a 100 kHz capacitance measuring assembly, which shall be capable of measuring capacitance from 10 pF to 150 pF with an accuracy of ±0.1 percent, so that the adjustable screw to which the inner conductor of the cable is connected, is positioned directly over the left hand terminal of the crystal socket. Set the pre-set input level potentiometer to its full counter clockwise position making note of its previous position. Set the contact point of the adjustable screw on the jig to as near contact with the left hand terminal of the crystal socket as can be determined visually.

(b) Set load capacity dial to "ZERO" (in line with vernier scale zero).

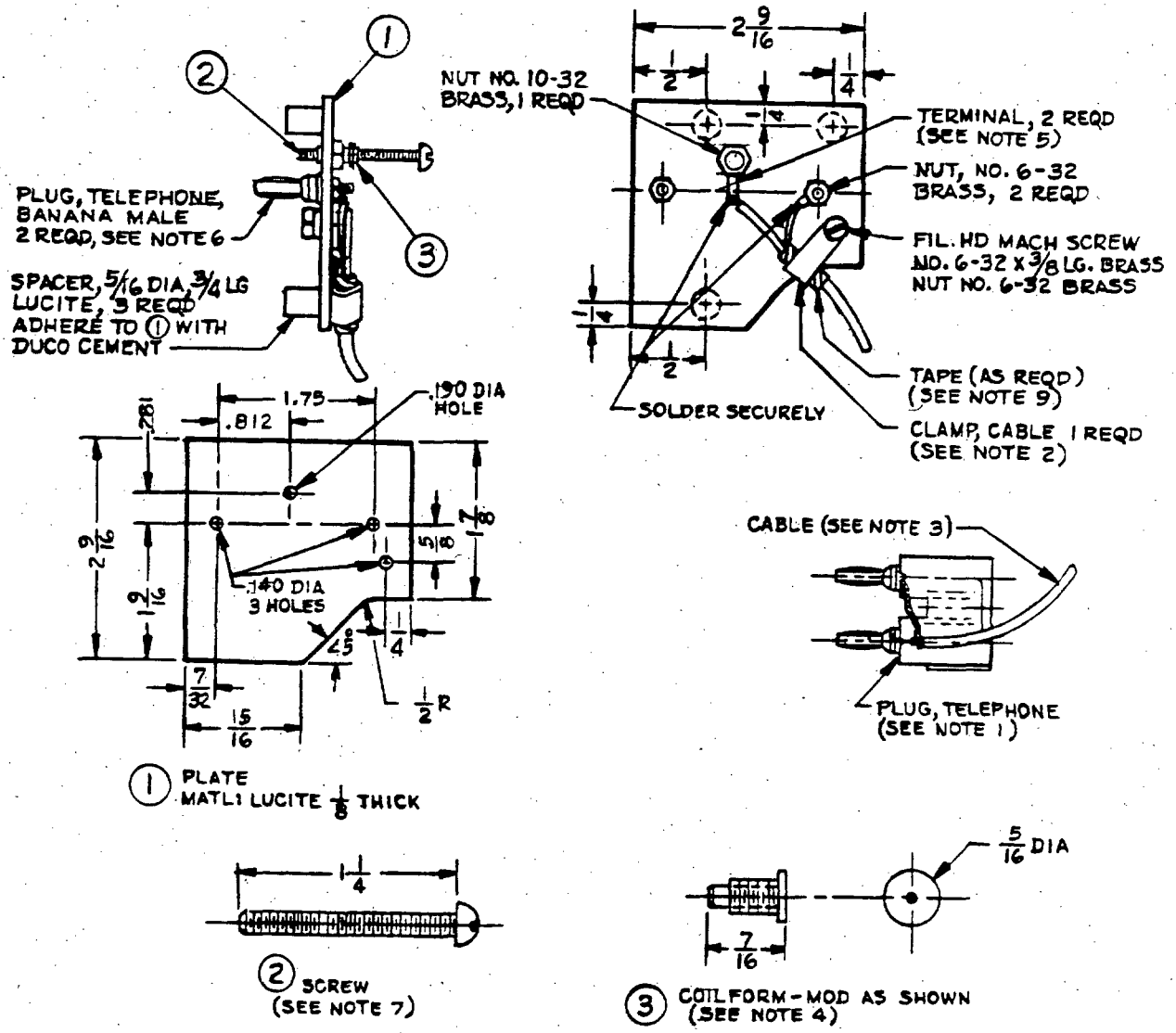
(c) Adjust the capacitance measuring assembly controls to obtain a null.

(d) Turn in the adjustable screw on the jig until it just makes contact with the crystal socket left hand terminal as indicated by loss of a null.

(e) Carefully back off the adjustable screw one turn to break contact.

(f) Readjust the capacitance measuring assembly controls to again obtain a null, using sufficient sensitivity to attain 0.1 percent accuracy. Record the capacitance reading.

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NOTE:

1. General Radio Type 274-MB or equal, West Concord.
2. H. H. Smith Type 102 or equal, Brooklyn.
3. RG-58C/U.
4. Cambridge Thermionics Type LS-7-C or equal, Cambridge.
5. Bud Co. Type TL-55 or equal, Cleveland.
6. Rich Co Plastic Type N-4 or equal.
7. STD MS-35214-47.
8. Tolerances, ±.002, ±1/64, ±3°.
9. Tape the Cable so that the Cable fits tight in the cable clamp.

FIGURE 1

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(g) Turn in the adjustable screw of the connecting screw until connection is again made, as indicated by the loss of the null and repeat (f). The difference in capacitance settings of (f) and (g) is the value of capacitance of the load capacitor at this dial setting.

(h) Repeat (g) at each dial division from 0 to 12, each 2 divisions from 12 to 30, and each 5 dial divisions from 30 to 100. The readings obtained shall meet the requirements of 3.7.7.

(i) Set: Frequency Range Kilocycles to "450 - 1100".

(j) Advance the Gain control fully clockwise.
Adjust the preset input level potentiometer for a 0.5 volt reading on the panel meter.

4.7.8 Capacitance across terminating resistance.

(a) Connect the capacitance measuring assembly unknown terminals to the "EXT." VTVM jack and equipment ground, using 16 AWG bare tinned wire.

(b) Set: Termination resistor (ohms) to "10,000"
Termination resistor full scale volts to "0.5"
Gain to full counter clockwise.

(c) Adjust the 100 kHz capacitance measuring assembly to obtain a null, using sufficient sensitivity to attain 0.1 percent accuracy of the assembly.

(d) Remove the lead from the "EXT." VTVM jack, but leave it in the same physical location and repeat (c).

(e) The difference between readings of (c) and (d) is the capacitance across the terminating resistance. When measured it shall meet the requirements of 3.7.8.

4.7.9 Accessories.

4.7.9.1 Adapters. Check the continuity of adapters, HC-5/U and the octal socket. It shall meet the requirements of 3.7.9.1.

4.7.9.2 Calibration resistor assemblies. Using a resistance bridge, measure the value of each calibration resistor. When measured the resistors shall meet the requirements of 3.7.9.2.

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4.7.9.3 Calibration potentiometer assemblies. Using a resistance bridge measure the maximum value of calibration potentiometers and continuity of control down to zero. When measured the calibration potentiometer assemblies shall meet the requirements of 3.7.9.3.

4.8 Service conditions.

4.8.1 Fungus. The equipment, including accessories, in its normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.1.

4.8.2 Low temperature. The equipment,including accessories, in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperatures in Step 2 held for 48 hours and in Step 4 held for 2 hours, and shall meet the requirements of 3.8.2.

4.8.3 High temperature. The equipment, including accessories, in its normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with temperature in Step 4 held for 2 hours, and shall meet the requirements of 3.8.3.

4.8.4 Humidity. The equipment, including accessories, in its normal operating configuration shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810, and shall meet the requirements of 3.8.4.

4.8.5 Altitude. The equipment, including accessories, in its normal operating configuration shall be subjected to the altitude test specified in Procedure I, Method 500 of MIL-STD-810, with altitude in Step 3 held for 1 hour and shall meet the requirements of 3.8.5.

4.8.6 Shock, bench-handling. The equipment, less accessories and case, shall be positioned as for servicing on a solid two inch fir bench top and shall be subjected to the following:

(a) Tilt up the equipment through an angle of 30 degrees, using one edge of the equipment as an axis. and permit the assembly to drop back freely to the horizontal, Repeat, using other practicable edges of the same horizontal face as axis, for a total of four drops on horizontal face.

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(b) Repeat (a) with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the equipment could be placed practicably during servicing.

(c) The equipment shall meet the requirements of 3.8.6.

4.8.7 Vibration. The equipment, including accessories, in its normal operating configuration shall be subjected to the vibration test as follows:

(a) Secure the equipment, less the case, directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured.

(b) The equipment shall be vibrated with the frequency varying between 10 and 55 Hz at a total excursion of 0.03 ± 0.006 inch, the frequency varying uniformly from 10 to 55 Hz and returning to 10 Hz in approximately 1 minute. Application of the vibration test specified above shall be in each of the following directions:

(1) Horizontal, parallel to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(2) Horizontal, at right angles to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(3) Vertically to the major horizontal axis of the equipment for a period of 15 minutes for the specified frequency range of 10 to 55 Hz.

(c) The equipment shall be inspected to ascertain that no fixed parts have become loose or damaged. The frequency range measurements per 3.7.1 and 4.7.1 shall be performed after completion of vibration test, and the equipment shall meet the requirements of 3.8.7.

4.8.8 Electromagnetic interference. The equipment shall meet the requirements of 3.8.8. All test setups and procedures employed shall conform to the measurement techniques of MIL-STD-462.

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4.9 Visual and mechanical. The complete equipment shall be examined for the defects listed in MIL-STD-252 and shall meet the requirements of 3.9.

4.10 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirements of 3.10. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, cover and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

(d) Location of crystal test holders, and terminals.

(e) Location of switches, sockets and fasteners.

(f) Location of cover and fasteners on carrying case which must mate.

(g) Location of panel mounting screw holes on basic unit for mating with case.

4.11 Burn-in and debugging. Each equipment, less accessories, produced shall be placed in its normal operating position on a test table with room temperature at 77°F ±5°F, with specified operating voltage applied from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period. In case of parts failure, the equipment shall be repaired, recalibrated, and the burn-in and debugging test repeated and shall meet the requirements of 3.11. A failed item analysis report shall be submitted on each part failure with contents of report per 5.10.5 of MIL-STD-781.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Test Set, Crystal Unit, Quartz TS-710( )/TSM shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Test Set, Crystal Unit, Quartz TS-710( )/TSM shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the method prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Electron tubes. All tubes not securely clamped in place shall be removed and with spare tubes shall be individually packaged in accordance with MIL-E-75.

5.1.1.4.3 Fuses. Fuses shall be packaged in accordance with Method III as follows: Cushion the fuses by wrapping together in paperboard conforming to PPP-P-291, type I. Secure wrap with gummed paper tape conforming to PPP-T-45, type III, grade A.

5.1.1.4.4 Lamp. Each lamp shall be packaged individually Method III as follows: Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, type I. Secure wrap with gummed paper tape conforming to PPP-T-45, type III.

5.1.1.4.5 Cord assembly CG-433A/U. Package the cord assembly method III. Wind the cord into a coil of proportionate dimensions and tie in three places with twine conforming to T-T-871.

5.1.1.4.6 Calibration chart. Package the calibration chart Method III. Place the chart between two pieces of fiberboard conforming to PPP-F-320, W6c, weather-resistant, designed as stiffeners. Secure the fiberboard with tape conforming to PPP-T-45, Type III.

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5.1.1.4.7 Accessory box with contents. Package each accessory box with contents Method III. Place the potentiometers, adapters and resistors assemblies within the designated compartments of the box, close cover and secure fastenings. Cushion the box by wrapping in paperboard conforming to PPP-P-291, Type III. Secure wrap with tape conforming to PPP-T-45, Type III.

5.1.1.4.8 Consolidated package. Place the items packaged as specified in 5.1.1.4.2 thru 5.1.1.4.7 together within a close fitting fiberboard box conforming to PPP-B-636, type CF, weather-resistant. Fill all voids with paperboard conforming to PPP-P-291, Type III, to prevent movement. Box closure shall be in accordance with the appendix of the box specification.

5.1.1.4.9 Test Set, Crystal Unit, Quartz TS-710( )/TSM. Each test set shall be packaged Method III as follows: Cushion each test set on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, W6c, weather-resistant, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item together with the consolidated package, packaged as specified in 5.1.1.4.8 within a close-fitting fiberboard box conforming to PPP-B-636, W5c, weather-resistant. Place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. Each Test Set, Crystal Unit, Quartz TS-710( )/TSM shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified. (See 6.2(b)).

5.2.1 Level A.

5.2.1.1 Palletized load. A quantity of containers packaged as specified in 5.1 shall be placed on a pallet, load type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, W5s, weather-resistant. The load shall be "banded" to the pallet by strapping.

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5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load. A quantity of items packaged as specified in 5.1 shall be packed within a close fitting fiberboard box conforming to PPP-B-640, class 2, grade A, style E, not to exceed 200 pounds or MIL-B-43291. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by non-metallic strapping or pressure sensitive filament tape banding conforming to PPP-S-760, type II or PPP-T-97, type IV respectively, selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B. Each Test Set, Crystal Unit, Quartz TS-710( )/TSM, packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C. Test Set, Crystal Unit, Quartz TS-710( )/TSM packaged as specified in 5.1 shall be packed as specified in 5.2.1, except that pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Test Set, Crystal Unit, Quartz TS-710( )/TSM is used to measure the frequency of crystals in depots and manufacturing plants.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment (Levels A, B or C) (see Section 5).

(c) If rough handling and functional tests are required.

(d) Quantity of first articles required

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(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

(h) Marking for shipment and shipping containers.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12F/U or AN/URM-73C.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (see 4.5.3.3)

6.5 Temperature conversion. Temperature conversion from fahrenheit to centigrade as listed in this specification are as follows:

160°F	=	71°C
125°F	=	51.5°C
77°F	=	25°C
-4°F	=	-20°C
-65°F	=	-53.9°C

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodians:
Army - EL

Preparing activity
Army - EL

Project Number 6625-A355

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-T-12333C(EL) TEST SET, CRYSTAL UNIT, QUARTZ TS-710( )/TSM

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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